

Result	Expected behavior	File
Passed	Approximately 70g removed, then 70g returned	gr.csv_50
Passed	Approximately 500g removed, then 500g returned	gr.csv_500
Passed	Approximately 1000g removed, then 1000g returned	gr.csv_1000

Why it matters	What it checks	Test
Prevents false events during normal operation.	No events are detected when the shelf remains at 500g	Constant weight
Confirms that insignificant changes below the threshold are rejected.	A 3g stable change is ignored	Small change
Prevents a bump or temporary disturbance from becoming an event.	A short jump to 550g that returns to 500g is ignored	Temporary spike
Confirms removal detection.	A sustained change from 500g to 0g creates one negative event	Single removal
Confirms addition detection.	A sustained change from 0g to 500g creates one positive event	Single addition
Prevents an invalid median calculation.	An empty list raises <code>ValueError</code>	Empty median
Confirms that at least two readings are required.	One measurement raises <code>ValueError</code>	Too few measurements
Verifies the plateau decision.	A quiet window is stable, while a 40g range is unstable	Stability check
Detects incorrectly structured files.	A CSV without the required columns raises <code>ValueError</code>	Missing columns
Prevents invalid arithmetic.	Text in a numeric column raises <code>ValueError</code>	Non-numeric value
Prevents undefined numerical behavior.	NaN is rejected	Non-finite value
Protects chronological event timing.	Duplicate timestamps raise <code>ValueError</code>	Repeated time